

CV - Melih Gümüş

Personal data

Surname	Gümüş
First name	Melih
Birth date	7 August 1996
Place of birth	Seyhan, Türkiye
Gender	Male
Nationality	Turkish
Mailing Address (flat number, street, No., ZIP code, city, state)	Emek Apartment, No. 2, 586th Street, Şeyh Şamil Neighbourhood, 06824 Ankara, Türkiye
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School education

FROM 09/2002 TO 06/2010	Türkkonut Elementry and Middle School / Ankara / Turkey
FROM 09/2010 TO 06/2014	Ufuk Arslan High School, Ankara, Türkiye
06/2014 Date of school-leaving qualification: 13 June 2014.	High School Diploma, Ufuk Arslan High School, Ankara, Türkiye.

Studies

FROM 09/2014 TO 07/2020 Çankaya University, Faculty of Engineering, Ankara, Türkiye.	Studies: - Bachelor's programme in Civil Engineering. - Language of instruction and examination: English. - The programme included one academic year of English preparatory education.
FROM 09/2014 TO 07/2020 Bachelor's Degree in Civil Engineering, Çankaya University, Ankara, Türkiye.	First university degree: - Graduation date: 3 July 2020. - Final cumulative GPA: 2.06/4.00.
FROM 09/2021 TO 07/2025 Ankara University, Faculty of Science, Ankara, Türkiye. Bachelor's programme in Physics.	Studies: - Major: Physics. - Language of instruction: Turkish. - Total programme workload: 240 ECTS.
07/2025 Bachelor of Science in Physics, Ankara University, Ankara, Türkiye.	Second university degree: - Graduation date: 8 July 2025. - Final cumulative GPA: 3.52/4.00. - Graduated with honours.

Other periods

FROM 09/2014 TO 06/2015 Successfully completed before commencing the English-medium Bachelor's programme in Civil Engineering.	English Preparatory School, Çankaya University, Ankara, Türkiye.
FROM 02/2025 TO 02/2025 3–7 February 2025.	Cosmology and Mathematical Physics Winter School, Turkish Academy of Sciences, Ankara, Türkiye.
FROM 07/2025 TO <i>present</i>	Independent study and research in theoretical physics, gravitation, differential geometry and teleparallel gravity.

Publications and research projects

2025	On the Nature of Gravitation and Geometric Drift of Spacetime Authors: Melih Gümüş and B. B. Öner Online preprint; currently under review in <i>Annals of Physics</i>. The study investigates the nature of gravitation through geometric drift vectors in a teleparallel framework.
2026	On the Geometric Interpretation of Non-Singular Black Hole Interiors Author: Melih Gümüş and B. B. Öner Published in <i>Modern Physics Letters A</i>. Article number: 2650184. The work investigates nonsingular black-hole interiors within teleparallel gravity. It uses geometric drift vectors and interprets torsion, rather than curvature, as the main geometric effect. Nonlinear contributions regularise the central region into a de Sitter-like core.
2026	On a Geometric Origin of Electromagnetism in Teleparallel Spacetime Author: Melih Gümüş Manuscript under editorial consideration. The work proposes a geometric interpretation of electromagnetism in teleparallel spacetime. Electromagnetic fields are represented through components of the torsion associated with the temporal tetrad. The framework reproduces Maxwell's equations, the Lorentz force and the Aharonov–Bohm phase in the appropriate limits.
TÜBİTAK 2209-A	Investigation of Liénard–Wiechert Potentials and Their Applications Undergraduate Research Project supported by TÜBİTAK (The Scientific and Technological Research Council of Türkiye). The project investigated the extension of Liénard–Wiechert potentials to QCD bound states by using the Cornell potential and Lorentz transformations. A theoretical framework was developed for heavy quarkonium moving in a quark–gluon plasma medium, with emphasis on relativistic electrodynamics, QCD-inspired modelling and bound-state dynamics.

Knowledge of languages

Month/year	Language	Language test / result / level
Current	Turkish	Mother tongue
Current	English	English-medium Bachelor's degree in Civil Engineering, including one academic year of English preparatory study. Self-assessed skills: - Listening B2; - Reading C1; - Writing B1; - Spoken Production B1; - Spoken Interaction B2.
06/2020	German	- University course: German I. - Course grade: BA.